MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

AMPLIFIER, AGC, ANALOG, KEY CODE DPA

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic amplifier module with an automatic gain control (AGC) function.

* 1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 2.22 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be DPA.

3.5.2 Size. The module size shall be 1B.

3.5.3 Weight. The module shall have a maximum weight of 50 grams.

3.5.4 Contact pins. The module shall have 20 contact pins positioned A1 through A20. Contact pins A21 through A40 are optional.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/245-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage of ±15.0 ±1.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of ±16.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

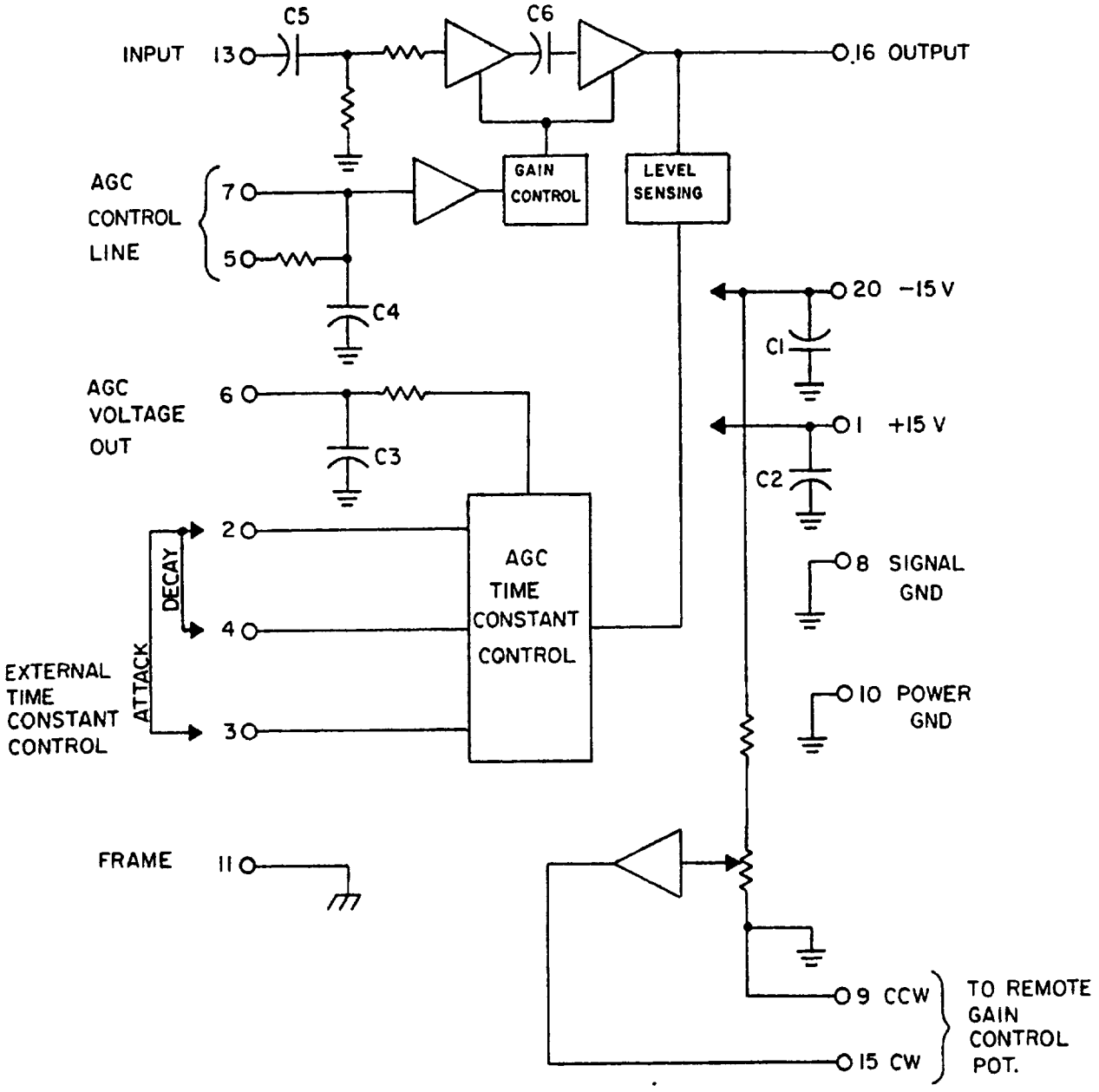
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
1. Pins 12, 14, 17, 18, and 19 are unused.
2. Pins 21-40 are optional.

FIGURE 1. DPA module function diagram, AGC amplifier.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	V_S stress	II	Q	-500	+500	-500	+500	-500	+500	-500	+500	mV
B1	Output short circuit current	III	S	1	40	1	40	1	40	1	40	mApk
B2	Output short circuit current	III	S	-1	-40	-1	-40	-1	-40	-1	-40	mApk
C1-C26	Isolation	IV	S	10	∞	10	∞	10	∞	10	∞	MΩ
D1, D2	Power supply current	V	S	0.8	15.0	0.8	15.0	0.8	15.0	0.8	15.0	mA
E	Maximum gain	VI	X	0.75	1.25	0.50	1.50	0.75	1.25	0.50	1.50	V rms
F	Minimum gain	VII	X	0.75	1.25	0.50	1.50	0.75	1.25	0.50	1.50	V rms
G1	AGC time constant, attack	VIII	X	2	6	2	6	2	7	2	8	s
G2	AGC time constant, attack	VIII	X	0	5	0	5	0	5	0	6	s
G3	AGC time constant, decay	VIII	X	5	11	5	11	4	12	4	12	s
G4	AGC time constant, decay	VIII	X	0	6	0	6	0	6	0	6	s
H	Supply voltage rejection	IX	S	0	100	0	100	0	100	0	100	mV
J1	Bandwidth	X	S	20	100	20	100	20	100	20	100	kHz
J2	Bandwidth	X	S	1	20	1	20	1	20	1	20	Hz
K1	Output voltage swing	XI	X	8.5	13.5	8.5	13.5	8.5	13.5	8.5	13.5	V
K2	Output voltage swing	XI	X	-8.5	-13.5	-8.5	-13.5	-8.5	-13.5	-8.5	-13.5	V
L	Noise	XII	X	0	100	0	100	0	100	0	100	mV rms
M1,M3	Capacitance	XIII	S	0.075	0.125	0.075	0.125	0.075	0.125	0.075	0.125	μF
M5	Capacitance	XIII	S	0.9	1.1	0.9	1.1	0.9	1.1	0.9	1.1	μF
M7	Capacitance	XIII	S	3,330	4,480	3,330	4,480	3,330	4,480	3,000	5,000	pF
M2,M4	Dissipation	XIII	S	0	50	0	50	0	50	0	50	%

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
M6,M8	Dissipation	XIII	S	0	5	0	5	0	5	0	5	%
N	Continuity	XIV	S	0	1	0	1	0	1	0	1	Ω
P	Distortion	XV	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	%
Q1,Q2	Input impedance check	XVI	S	10	∞	10	∞	10	∞	10	∞	kΩ
R1	Remote gain function	XVII	X	---	1	---	1	---	1.2	---	1.35	V rms
R2	Remote gain function	XVII	X	← See table XVII →								V rms
R3	Remote gain function	XVII	X	---	0.75	---	0.75	---	0.75	---	0.75	V rms
R4	Remote gain function	XVII	X	← See table XVII →								V rms

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through R shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence o tests, the order of testing shall be as follows: A, B, C where applicable, followed b the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through R in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through M and P through R shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through M and P through R shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through R shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

*6.3 Cross reference of classification. Part number M28787/245-1 supersedes part number M28787/245. The two parts are interchangeable.

*6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

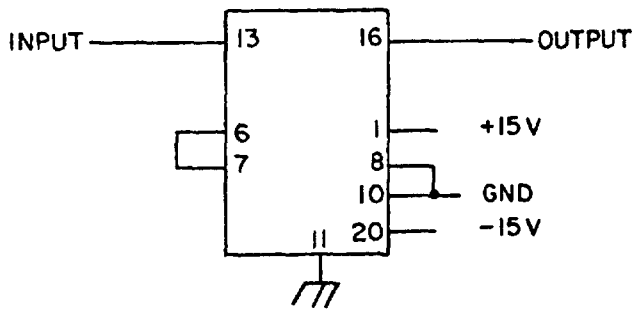
Maximum power dissipation	
+16.5 V dc supply	248 mW (No signal)
-16.5 V dc supply	248 mW (No signal)
Supply voltages	+15 V dc ±10 percent -15 V dc ±10 percent
Supply voltage rejection	20 dB
Controlled output level	1 V rms ±50%, f = 20 Hz to 20 kHz, $R_L \geq 5$ kΩ, input signal = 1 mV rms to 1 V rms

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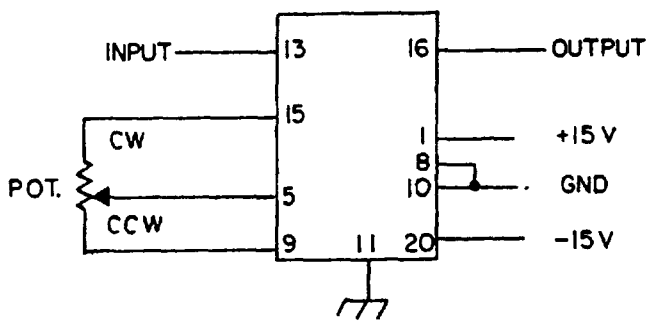
* 6.5 Module function. This module contains an audio amplifier with an automatic gain control function.

* 6.6 Module circuit connections. The module may be connected for normal AGC function or for remote gain control function.

6.6.1 Normal AGC function. Pins 6 and 7 are tied together for normal operation.



6.6.2 Remote gain control function. Amplifier gain is remote controlled from a potentiometer (100 kΩ ±10%, taper E, reverse log, RV6SAYSD104E, MIL-R-94). No audio signal appears on the wiring to the potentiometer, therefore, no shielding is required.



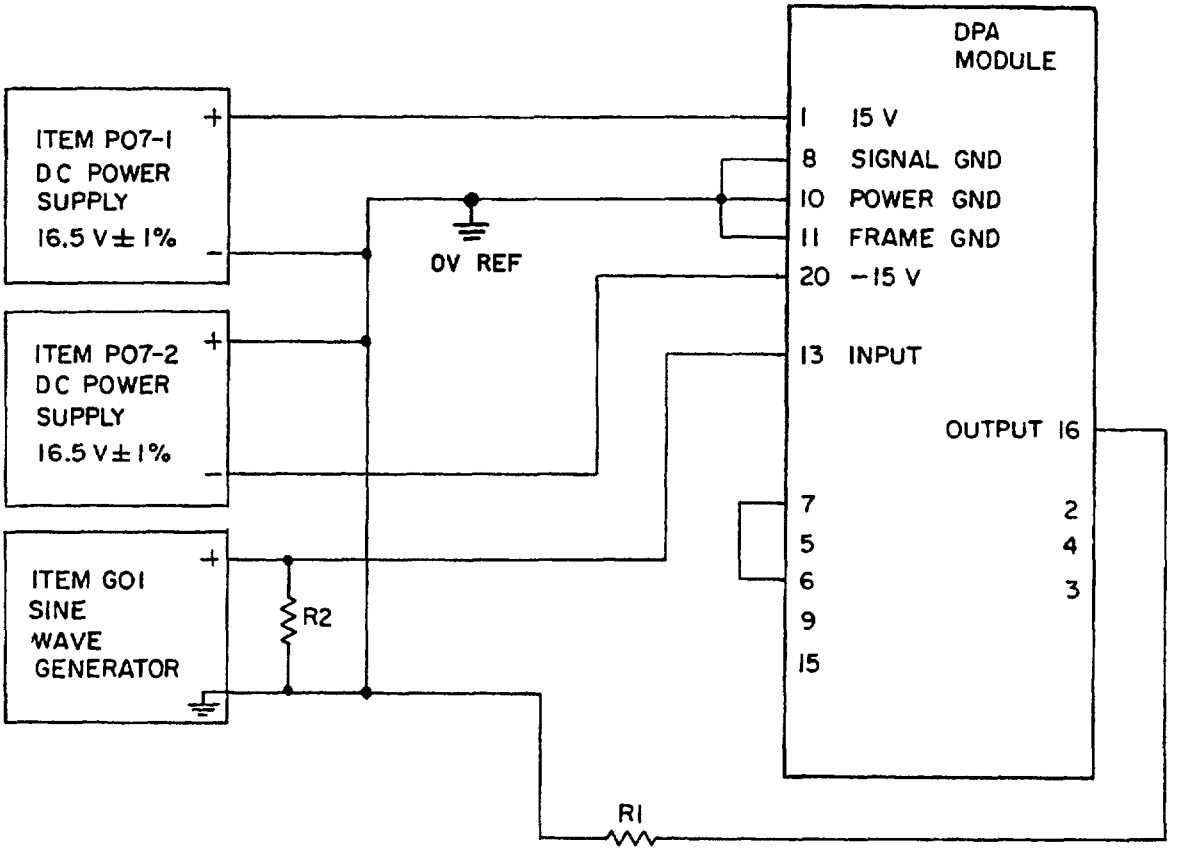
6.7 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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NOTES:
1. R1 = 5 kΩ ±1%, 1/4 W.
2. R2 = 51Ω ±5%, 1 W.
3 Item GO1 setup:
Amplitude = 1.0 ±0.1 V_{p-p},
Frequency = 1 kHz ±2%.

FIGURE 2. Operating temperature, component temperature, and life test circuit

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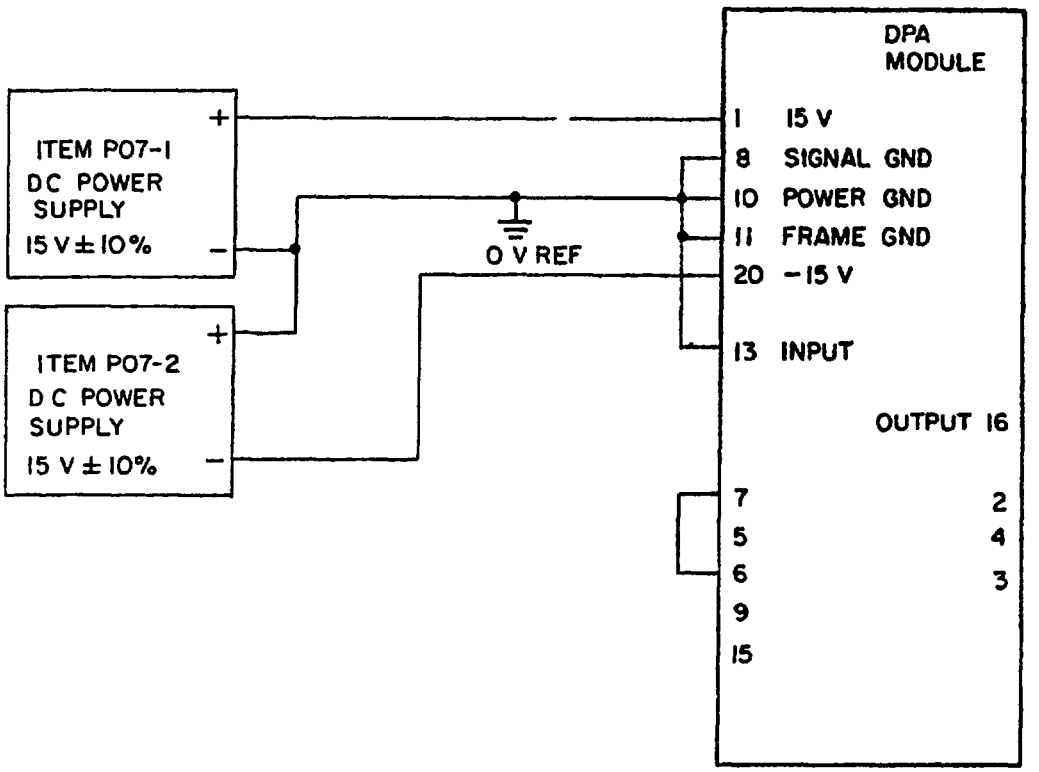


FIGURE 3. Vibration test circuit.

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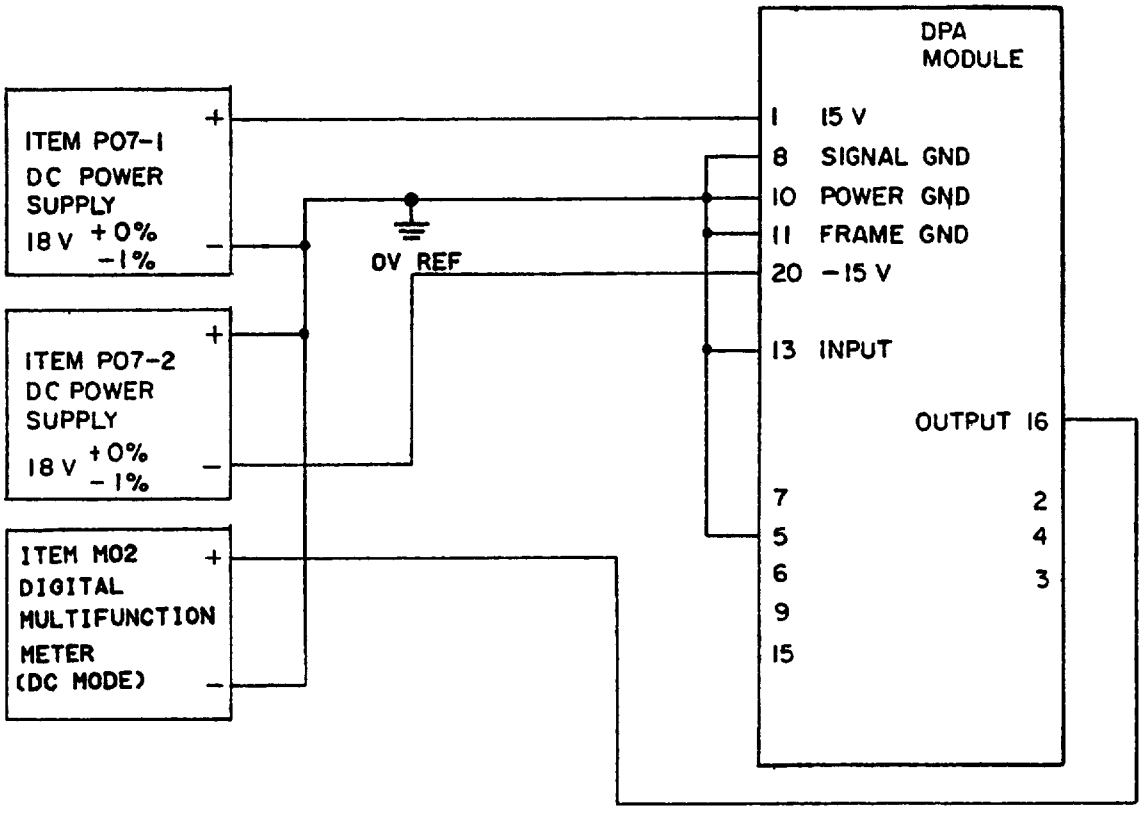
TABLE II. V_S stress test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
A	V_S stress	-500	+500	mV	4

TABLE III. Output short circuit current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
B1	Output short circuit current	1	40	mApk	5
B2	Output short circuit current	-1	-40	mApk	5

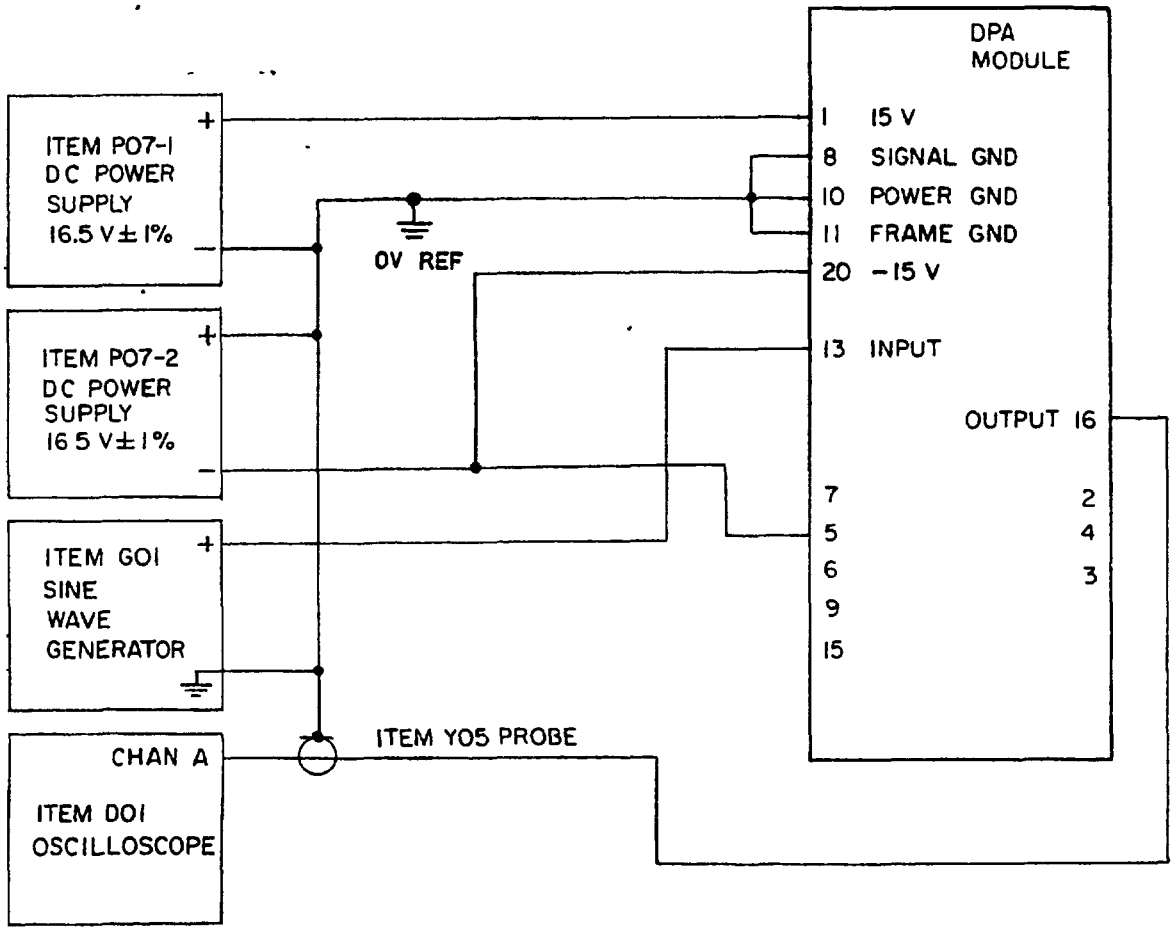
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NOTE: Measurement shall be recorded after module has been connected for a minimum of 1 minute.

* FIGURE 4. V_S stress test circuit for table II.

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NOTES:
1. When performing the output short circuit tests, the measurement shall be taken after 1 minute.
2. Item G01 setup:
   Amplitude = 1.0 ±0.1 V_{p-p}.
   Frequency = 1 kHz ±2%.
3. For test condition B1 measure from the 0 mA level to the positive peak of the output current waveform.
4. For test condition B2 measure from the 0 mA level to the negative peak of the output current waveform.

* FIGURE 5. Output short circuit current test circuit for table III.

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TABLE IV. Isolation test.

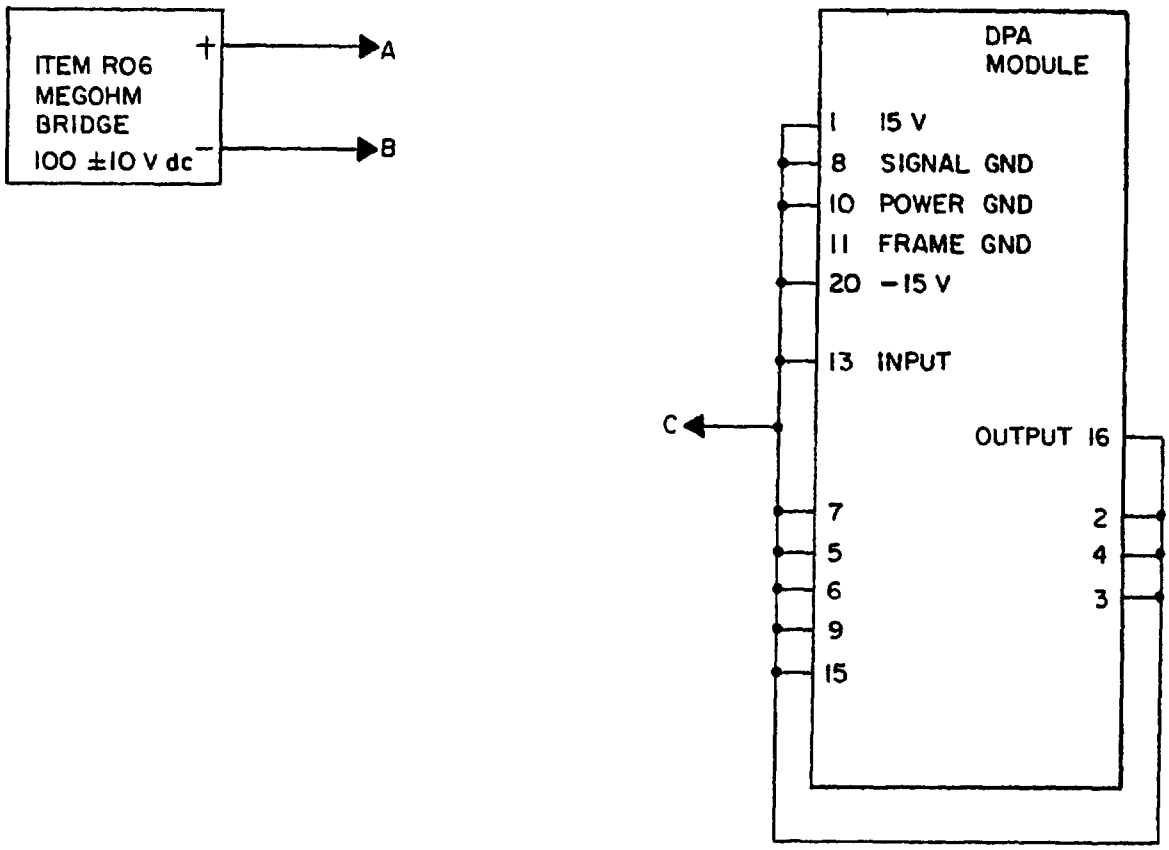
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	Isolation, frame to used pins	11	C	10	∞	MΩ	6
C2	Isolation, unused to used pins	12,14,17,18,19, 21 through 40	C	"	"	"	"
C3	Isolation, unused to unused pins	14,17,18,19, 21 through 40	12	"	"	"	"
C4	"	17,18,19, 21 through 40	14	"	"	"	"
C5	"	18,19, 21 through 40	17	"	"	"	"
C6	"	19, 21 through 40	18	"	"	"	"
C7	"	21 through 40	19	"	"	"	"
C8	"	22 through 40	21	"	"	"	"
C9	"	23 through 40	22	"	"	"	"
C10	"	24 through 40	23	"	"	"	"
C11	"	25 through 40	24	"	"	"	"
C12	"	26 through 40	25	"	"	"	"
C13	"	27 through 40	26	"	"	"	"
C14	"	28 through 40	27	"	"	"	"
C15	"	29 through 40	28	"	"	"	"
C16	"	30 through 40	29	"	"	"	"
C17	"	31 through 40	30	"	"	"	"
C18	"	32 through 40	31	"	"	"	"
C19	"	33 through 40	32	"	"	"	"
C20	"	34 through 40	33	"	"	"	"

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TABLE IV. Isolation test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C21	Isolation, unused to unused pins	35 through 40	34	10	∞	MΩ	6
C22	"	36 through 40	35	"	"	"	"
C23	"	37 through 40	36	"	"	"	"
C24	"	38 through 40	37	"	"	"	"
C25	"	39,40	38	"	"	"	"
C26	"	40	39	"	"	"	"

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

FIGURE 6. Isolation test circuit for table IV.

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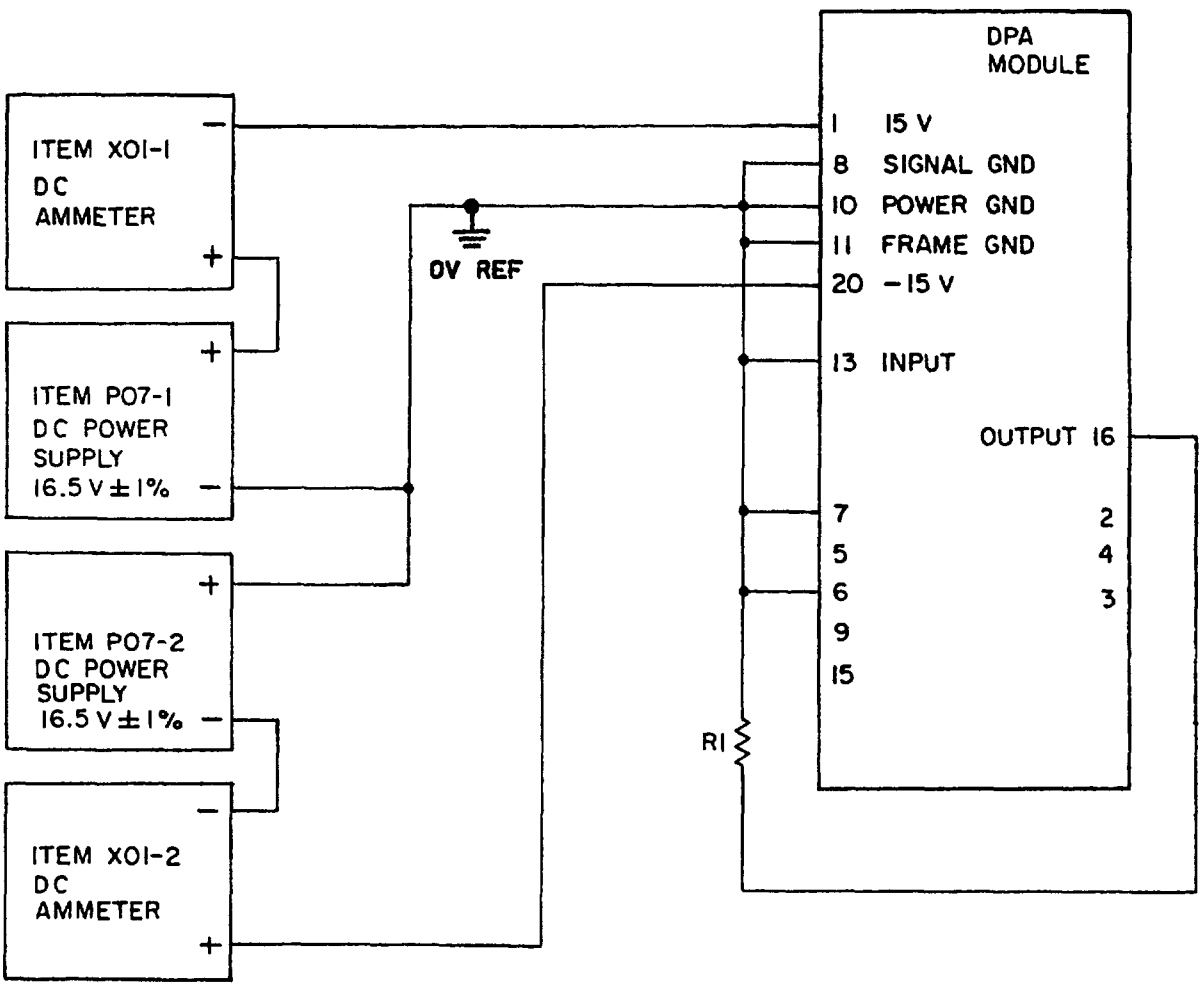
TABLE V. Power supply current test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
D1	Power supply current, positive	0.8	15.0	mA	7
D2	Power supply current, negative	0.8	15.0	mA	7

TABLE VI. Maximum gain test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
E	Maximum gain	0.75	1.25	V rms	8

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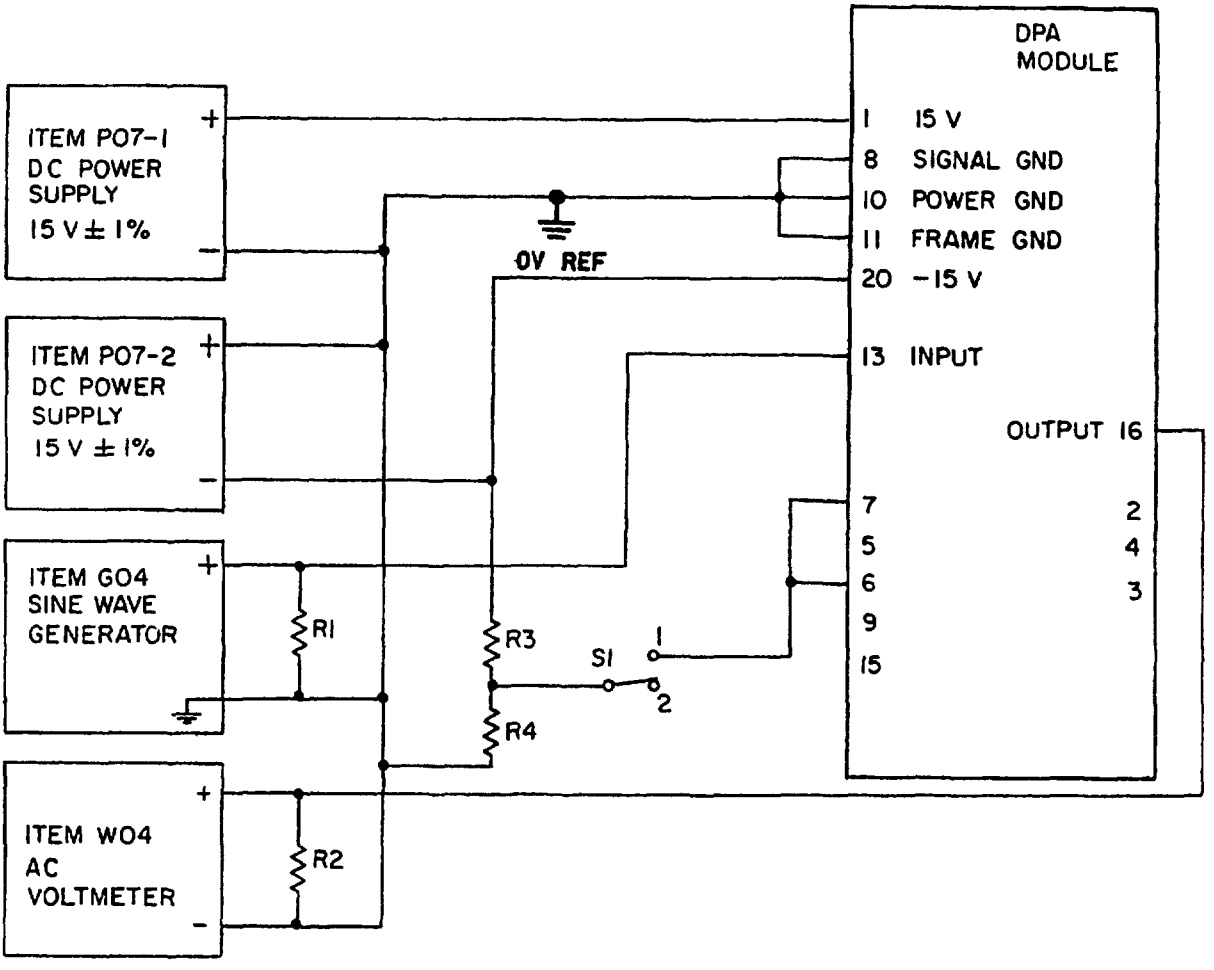
CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

NOTE: R1 = 5 kΩ ±1%, 1/4 W.

FIGURE 7. Power supply current test circuit for table V.

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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 5 kΩ ±1%, 1/4 W.
3. R3 = 11.5 kΩ ±1%, 1/4 W.
4. R4 = 10 kΩ ±1%, 1/4 W.
5. S1 = SPDT switch.
6. Adjust item G04 for an input of 1 mV rms ±1%, 1 kHz ±5% at pin 13 (use item W04).
7. Switch S1 is used to shorten test time due to the long time constant of AGC action. The switch is put into position 1 for about 1 second then put into position 2. Measurements are taken when the module output voltage stops changing.

* FIGURE 8. Maximum gain test circuit for table VI.

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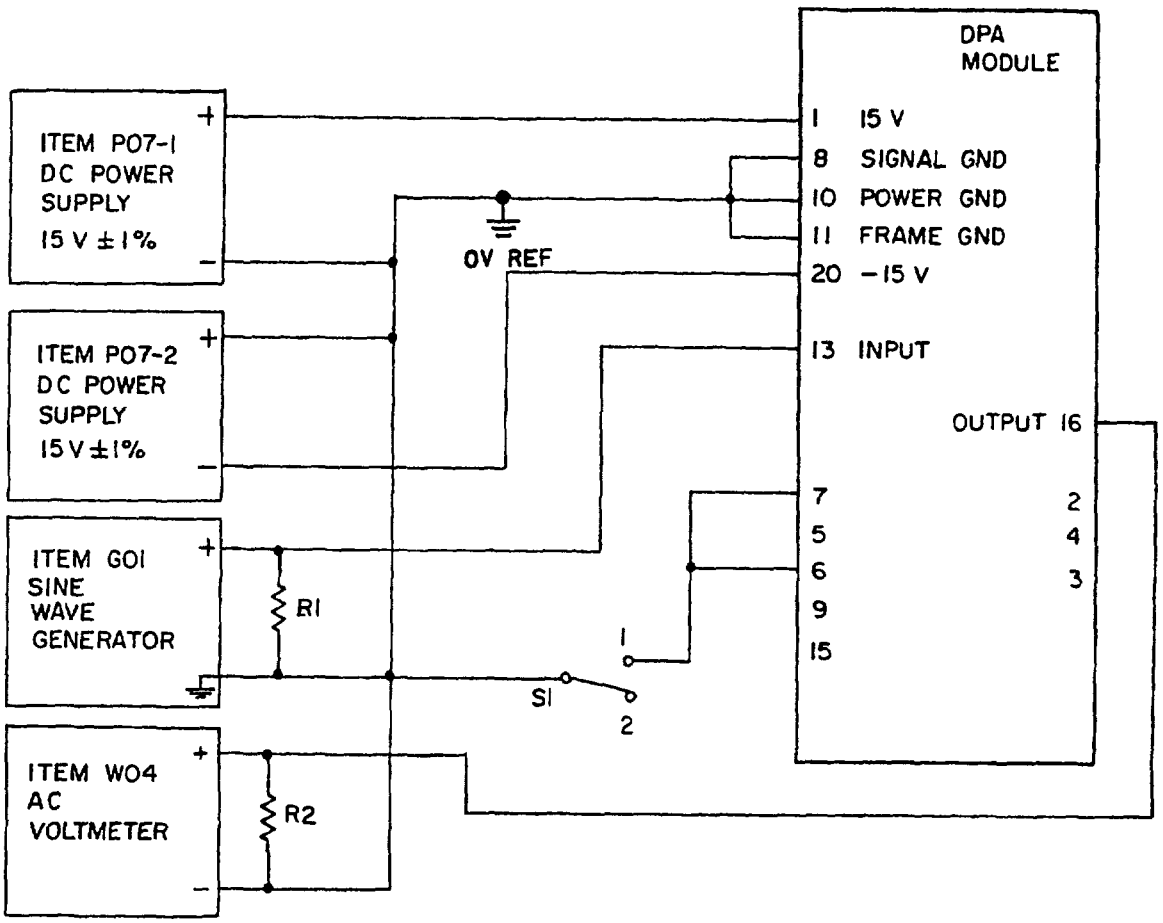
TABLE VII. Minimum gain test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
F	Minimum gain	0.75	1.25	V rms	9

TABLE VIII. AGC time constant test.

Test condition	Parameter	S1 position	S2 position	Initial limits 25°C ±5°C ambient		Unit	Test figure
				Min	Max		
G1	AGC time constant, attack	Start at 2, move to 1	1	2	6	s	10
G2	AGC time constant, attack	Start at 2, move to 1	2	0	5	"	"
G3	AGC time constant, decay	Start at 1, move to 2	1	5	11	"	"
G4	AGC time constant, decay	Start at 1, move to 2	2	0	6	"	"

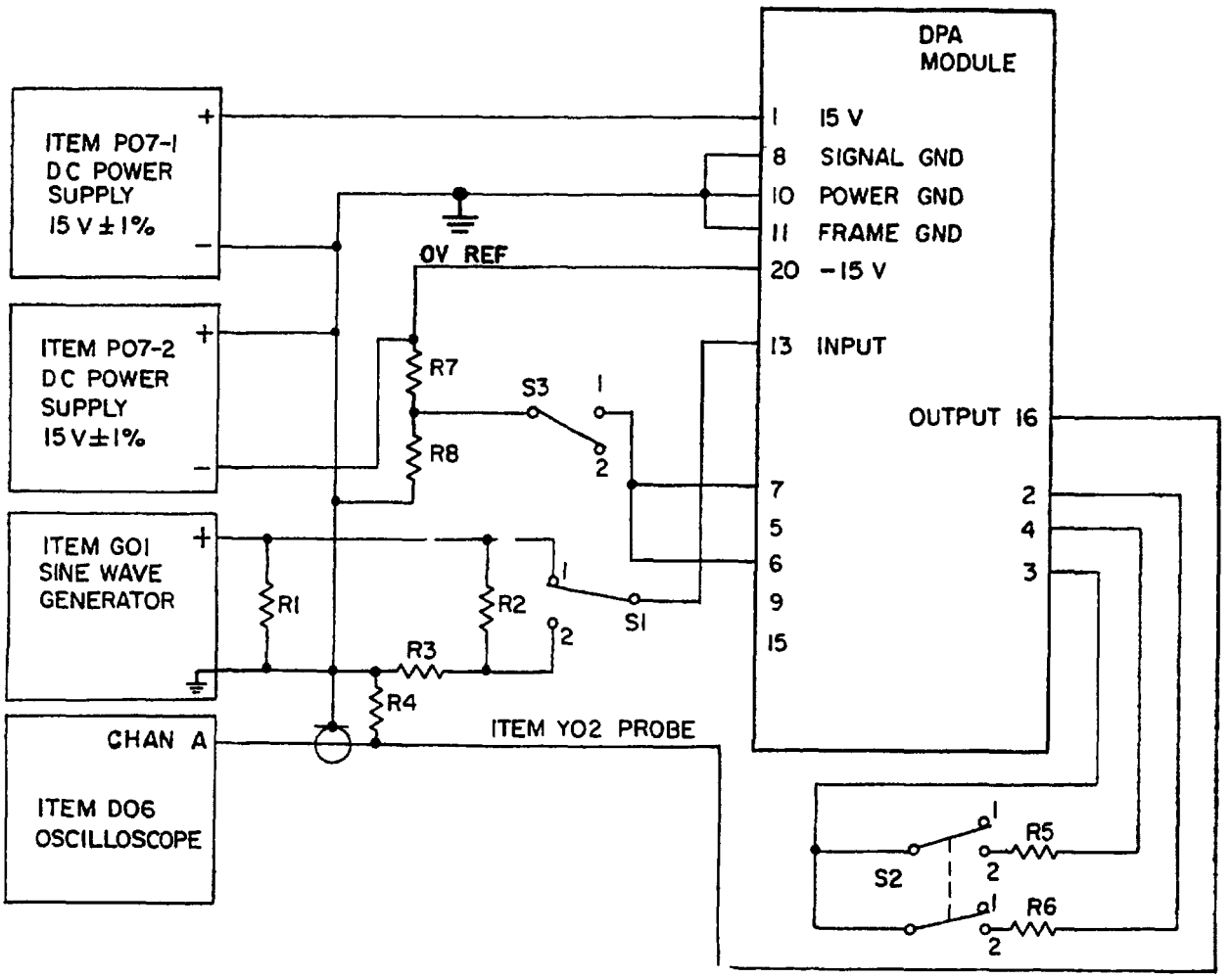
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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 5 kΩ ±1%, 1/4 W.
3. S1 = SPDT switch.
4. Adjust item G01 for an input of 1 V rms ±1%, 1 kHz ±5% at pin 13 (use item W04).
5. Switch S1 is used to shorten test time due to the long time constant of AGC action. The switch is put into position 1 for about 1 second then put into position 2. Measurements are taken when the module output voltage stops changing.

* FIGURE 9. Minimum gain test circuit for table VII.

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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 220Ω ±1%, 1/4 W.
3. R3 = 100Ω ±1%, 1/4 W.
4. R4 = 5 kΩ ±1%, 1/4 W.
5. R5 = 100 kΩ ±5%, 1/8 W, carbon.
6. R6 = 390 kΩ ±5%, 1/8 W, carbon.
7. R7 = 11.5 kΩ ±1%, 1/4 W.
8. R8 = 10 kΩ ±1%, 1/4 W.
9. S1 = S3 = SPDT switch.
10. S2 = DPDT switch.
11. Item G01 setup:
    Amplitude = 32 mV rms ±5%.
    Frequency = 1 kHz ±5%.
12. Test procedure:
    a. Switch S3 is used to shorten test time due to the long time constant of AGC action. The switch is put into position 1 for about 1 second then put into position 2.
    b. Move switch S1 to the starting position and allow output to stablize then read output level.
    c. Move switch S1 to other position and start timing.
    d. Stop timing when output returns to within 0.1 dB (1 percent) of starting value.

* FIGURE 10. AGC time constant test circuit for table VIII.

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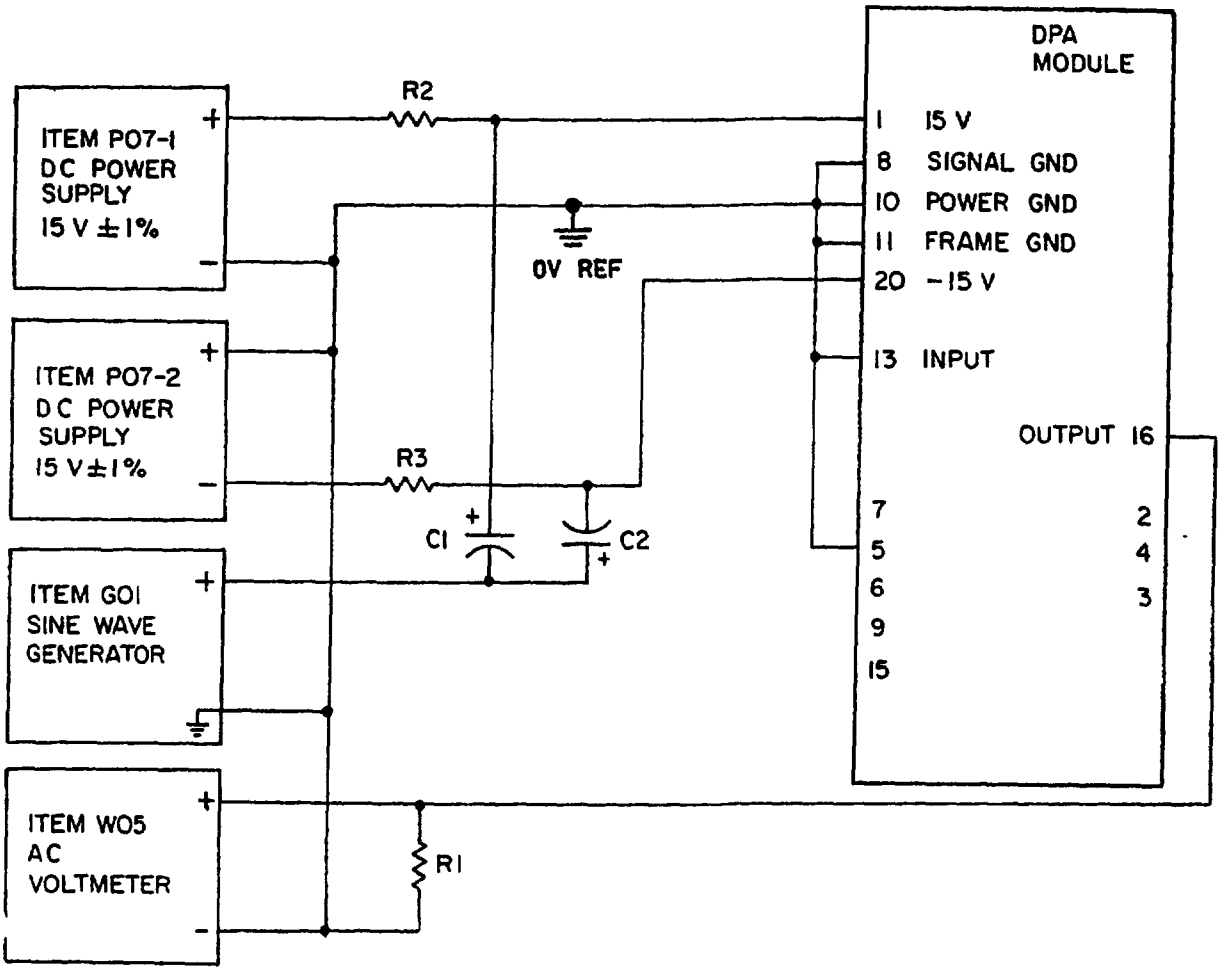
TABLE IX. Supply voltage rejection test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
· H	Supply voltage rejection	0	100	mV	11

TABLE X. Bandwidth test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
J1	Bandwidth	20	100	kHz	12
J2	Bandwidth	1	20	Hz	12

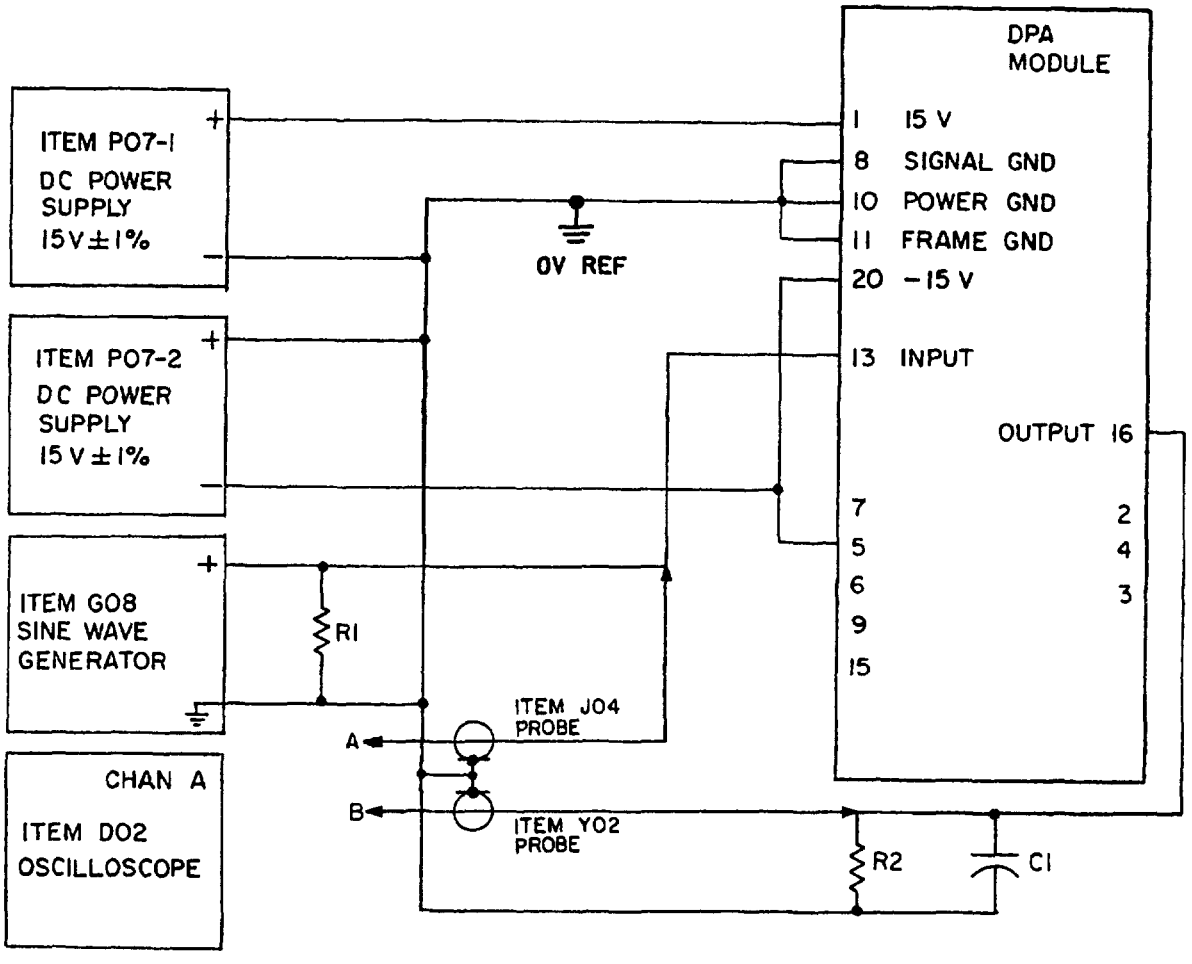
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NOTES:
1. R1 = 5 kΩ ±1%, 1/4 W.
2. R2 = R3 = 100Ω ±1%, 1/4 W.
3. C1 = C2 = 20 µF ±20%, 20 V, electrolytic.
4. Adjust item G01 for an input of 1 V rms ±5%, 1 kHz ±5% at pins 1 and 20 (use item W05).

* FIGURE 11. Supply voltage rejection test circuit for table IX.

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NOTES:
1. R1 = 600Ω ±5%, 1/4 W.
2. R2 = 5 kΩ ±1%, 1/4 W.
3. C1 = 100 pF ±20%.
4. Test procedure:

   a. Connect point B to channel A of item D02 and adjust item G08 for an output (pin 16) of 1 V rms and 1 kHz.
   b. Connect point A to channel A of item D02 and measure item G08 voltage amplitude. Maintain this amplitude within ±1% during test conditions J1 and J2.
   c. For test condition J1 connect point B to channel A of item D02 and continuously increase the frequency of item G08 until the voltage at pin 16 decreases to 0.7 V rms (-3 dB point). The item G08 frequency at the -3 dB point shall conform to table X requirements.
   d. For test condition J2 continuously decrease item G08 frequency until the voltage at pin 16 decreases to 0.7 V rms (-3 dB point). The item G08 frequency at this -3 dB point shall conform to table X requirements.

* FIGURE 12. Bandwidth test circuit for table X.

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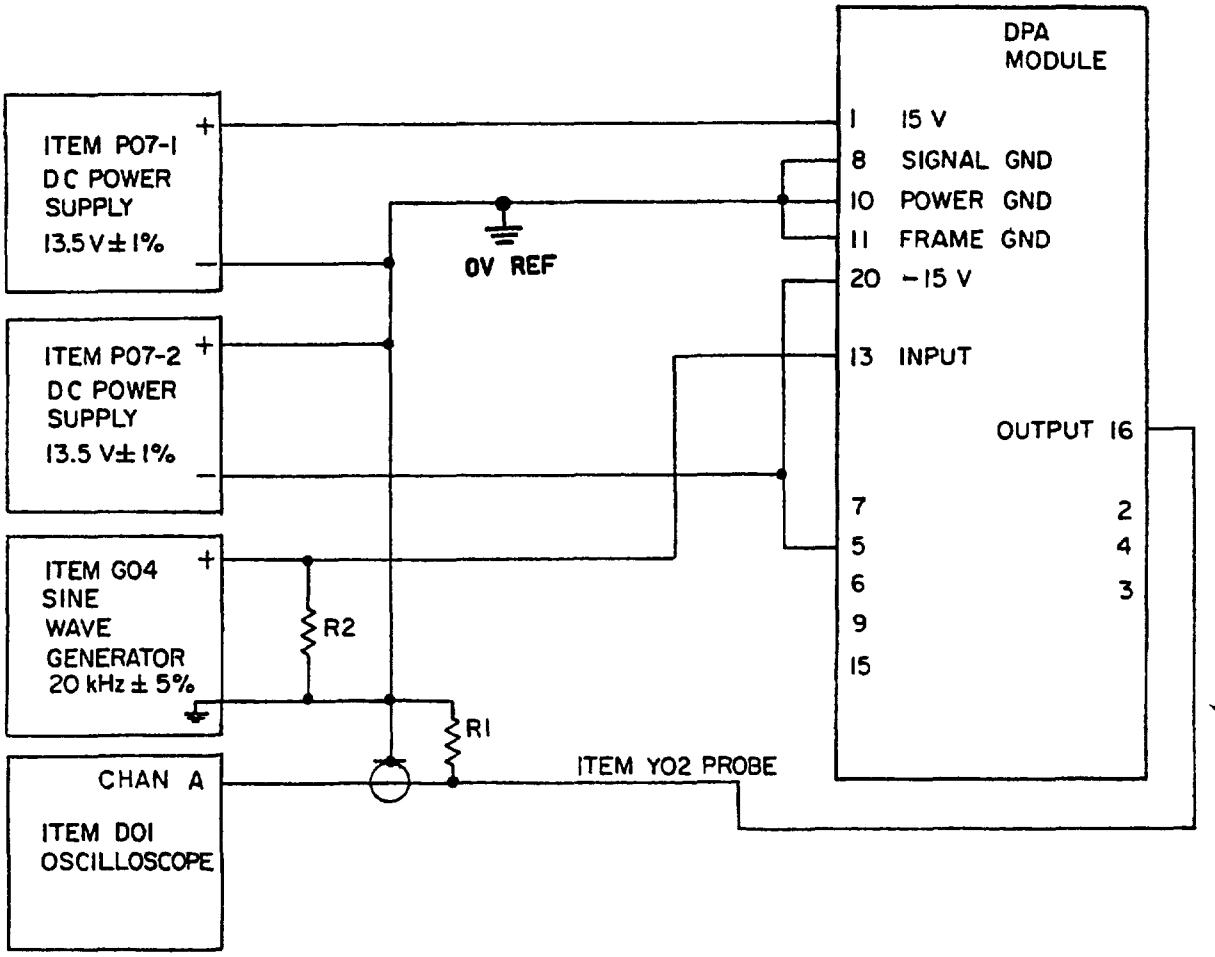
TABLE XI. Output voltage swing test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
K1	Output voltage swing	8.5	13.5	V	13
K2	Output voltage swing	-8.5	-13.5	V	13

TABLE XII. Noise test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
L	Noise	0	100	mV rms	14

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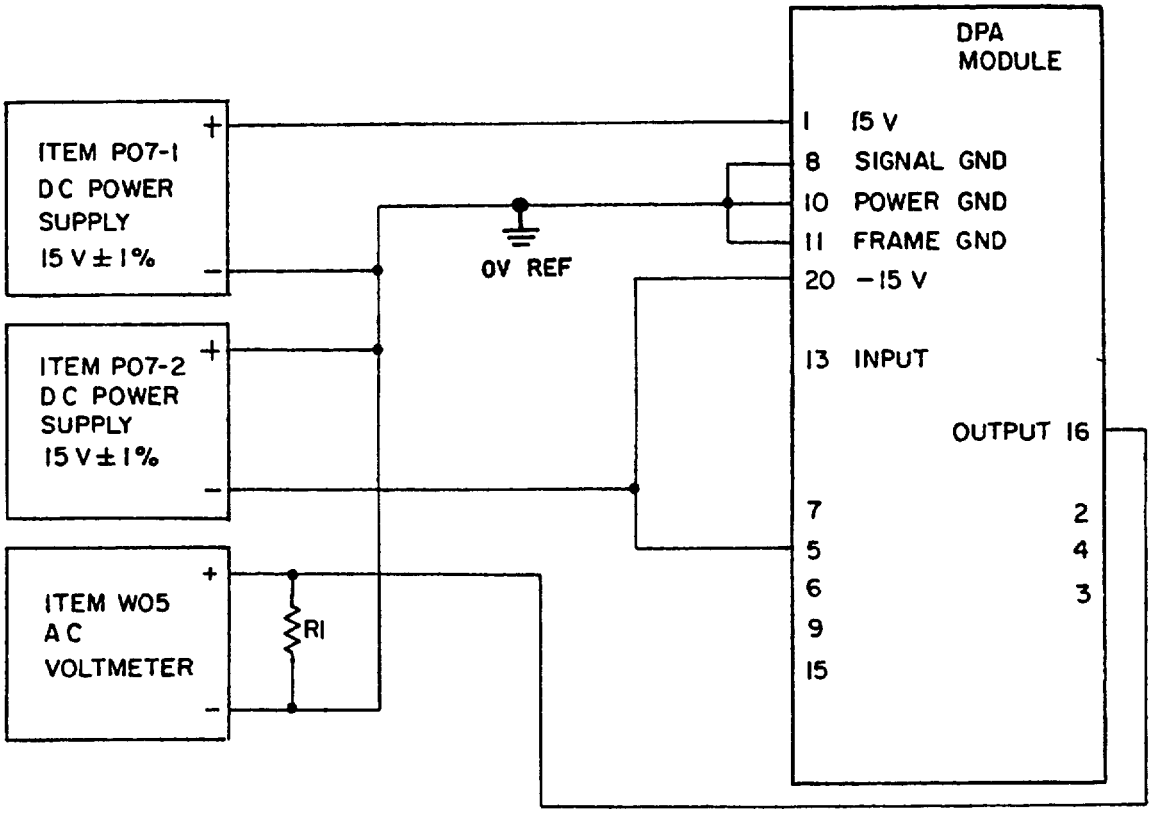


NOTES:
1. R1 = 5 kΩ ±1%, 1/4 W.
2. R2 = 51Ω ±5%, 1 W.
3. Test procedure:

   a. Monitor the waveform at pin 16 while increasing item G04 amplitude until clipping occurs on both the positive and negative peaks.
   b. For test condition K1, the clipping point shall be measured on the positive side of the voltage swing.
   c. For test condition K2, the clipping point shall be measured on the negative side of the voltage swing.

* FIGURE 13. Output voltage swing test circuit for table XI.

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NOTE: R1 = 5 kΩ ±1%, 1/4 W.

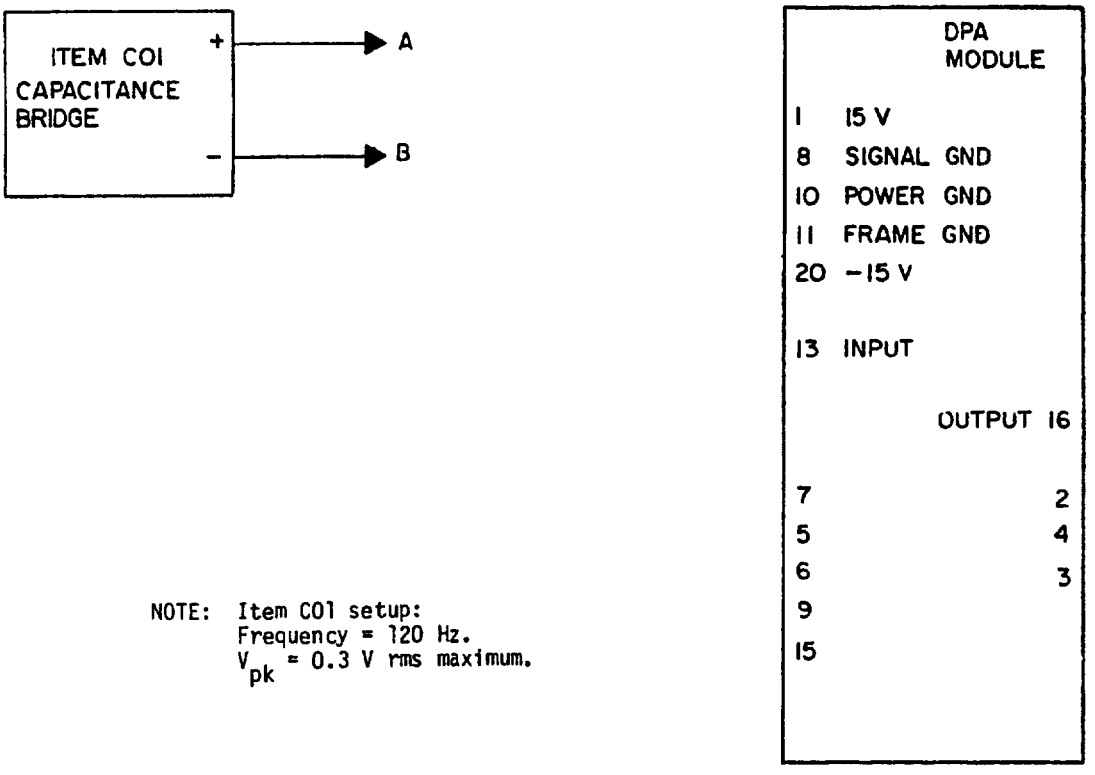
* FIGURE 14. Noise test circuit for table XII.

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TABLE XIII. Capacitance and dissipation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	C1 capacitance	20	8	0.075	0.125	µF	15
M2	C1 dissipatior	20	"	0	50	%	"
M3	C2 capacitance	1	"	0.075	0.125	µF	"
M4	C2 dissipation	1	"	0	50	%	"
M5	C3 capacitance	6	"	0.9	1.1	µF	"
M6	C3 dissipation	6	"	0	5	%	"
M7	C4 capacitance	7	"	3,330	4,480	pF	"
M8	C4 dissipation	7	"	0	5	%	"

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NOTE: Item COI setup:
Frequency = 120 Hz.
V_{pk} = 0.3 V rms maximum.

* FIGURE 15. Capacitance and dissipation test circuit for table XIII.

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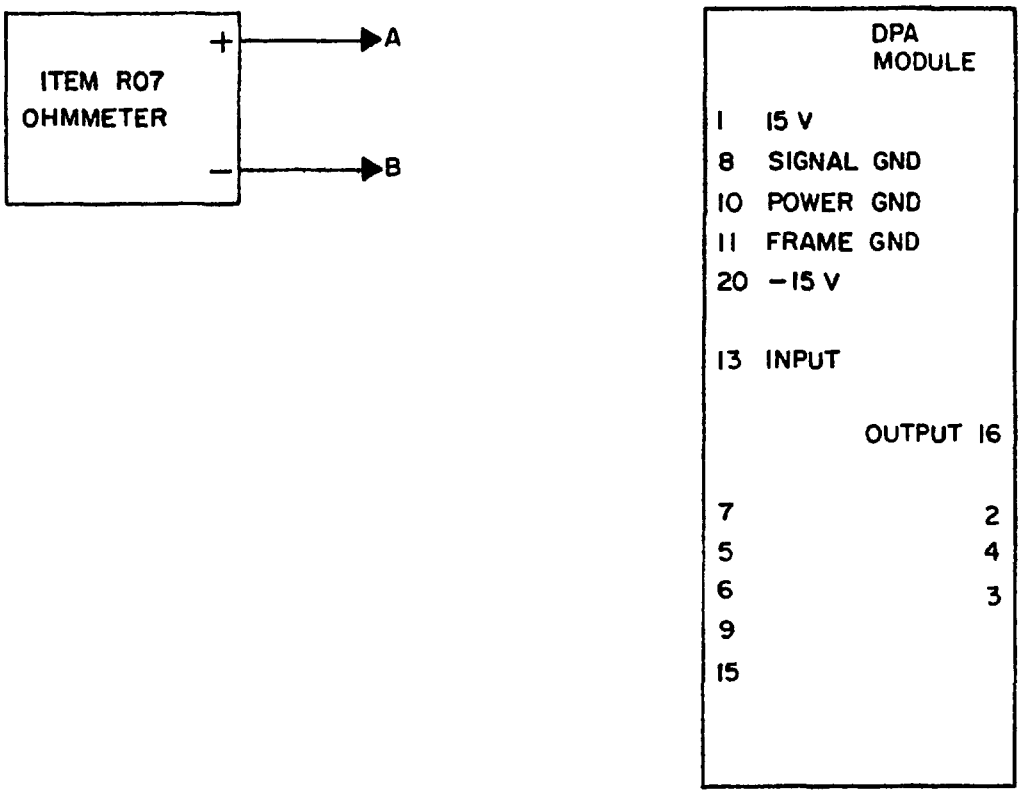
TABLE XIV. Continuity test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
N	Continuity	11	Frame	0	1	Ω	16

TABLE XV. Distortion test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
P	Distortion	0.0	1.5	%	17

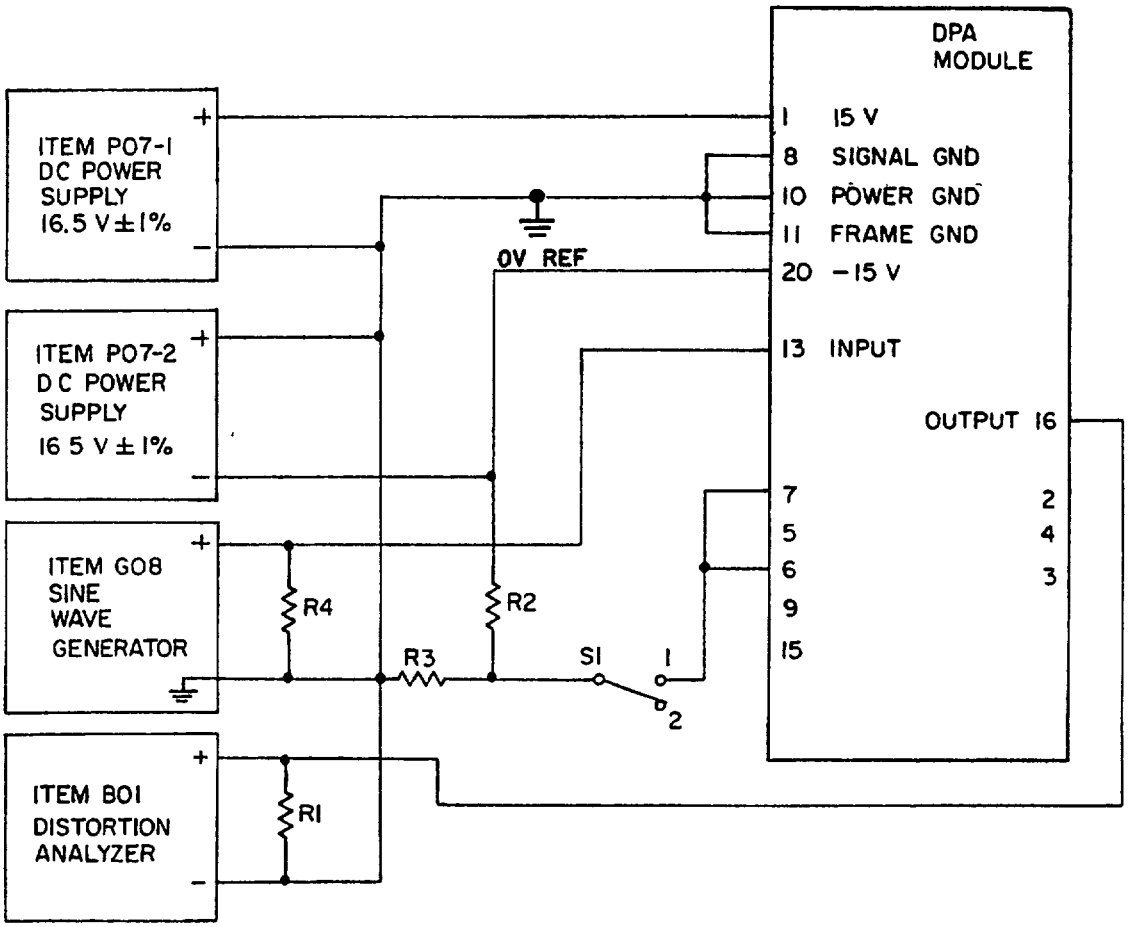
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NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

* FIGURE 16. Continuity test circuit for table XIV.

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NOTES:
1. R1 = 5 kΩ ±1%, 1/4 W.
2. R2 = 11.5 kΩ ±1%, 1/4 W.
3. R3 = 10 kΩ ±1%, 1/4 W.
4. R4 = 600Ω ±5%, 1/4 W.
5. S1 = SPDT switch.
6. Adjust item G08 for an input of 2 kHz ±5% and 100 mV rms ±1% at pin 13.
7. Switch S1 is used to shorten test time due to the long time constant of AGC action. The switch is put into position 1 for about 1 second then put into position 2. Measurements are taken when the module output voltage stops changing.

* FIGURE 17. Distortion test circuit for table XV.

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*TABLE XVI. Input impedance check test.

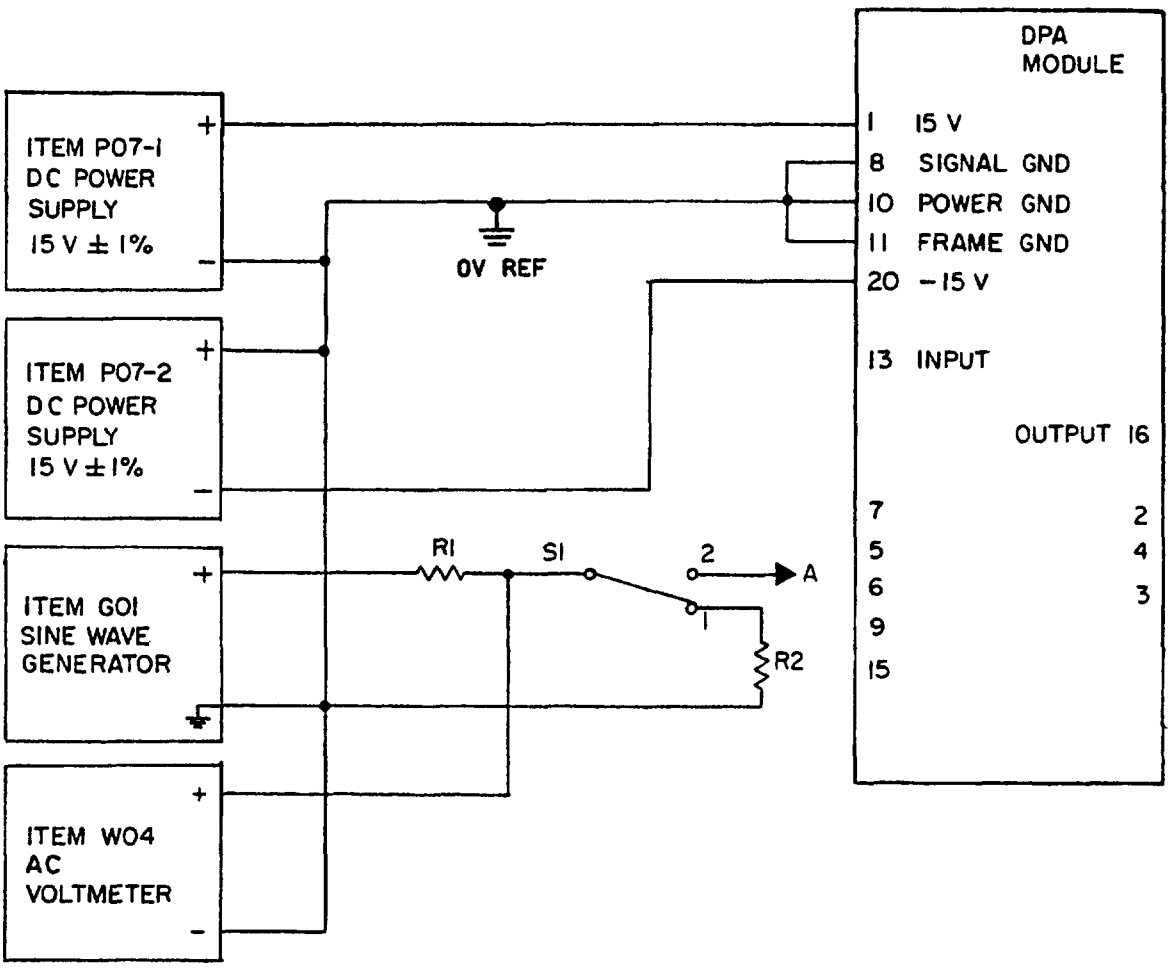
Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
Q1	Input impedance check	13	100	220	mV	18
Q2	Input impedance check	5	100	220	mV	18

$$\text{Table I limits} = \frac{1000(\text{Table XVI limits})}{0.22 - \text{Table XVI limits}} .$$

TABLE XVII. Remote gain function test.

Test condition	Parameter	S1 position	Item G04 output ±1%	Record voltage	Initial limits 25°C ±5°C ambient		Unit	Test figure
					Min	Max		
R1	Remote gain function	1	1 V rms	V1=	---	1	V rms	19
R2	"	2	1 V rms	---	V1	---	"	"
R3	"	3	1 mV rms	V2=	---	0.75	"	"
R4	"	4	1 mV rms	---	V2	---	"	"

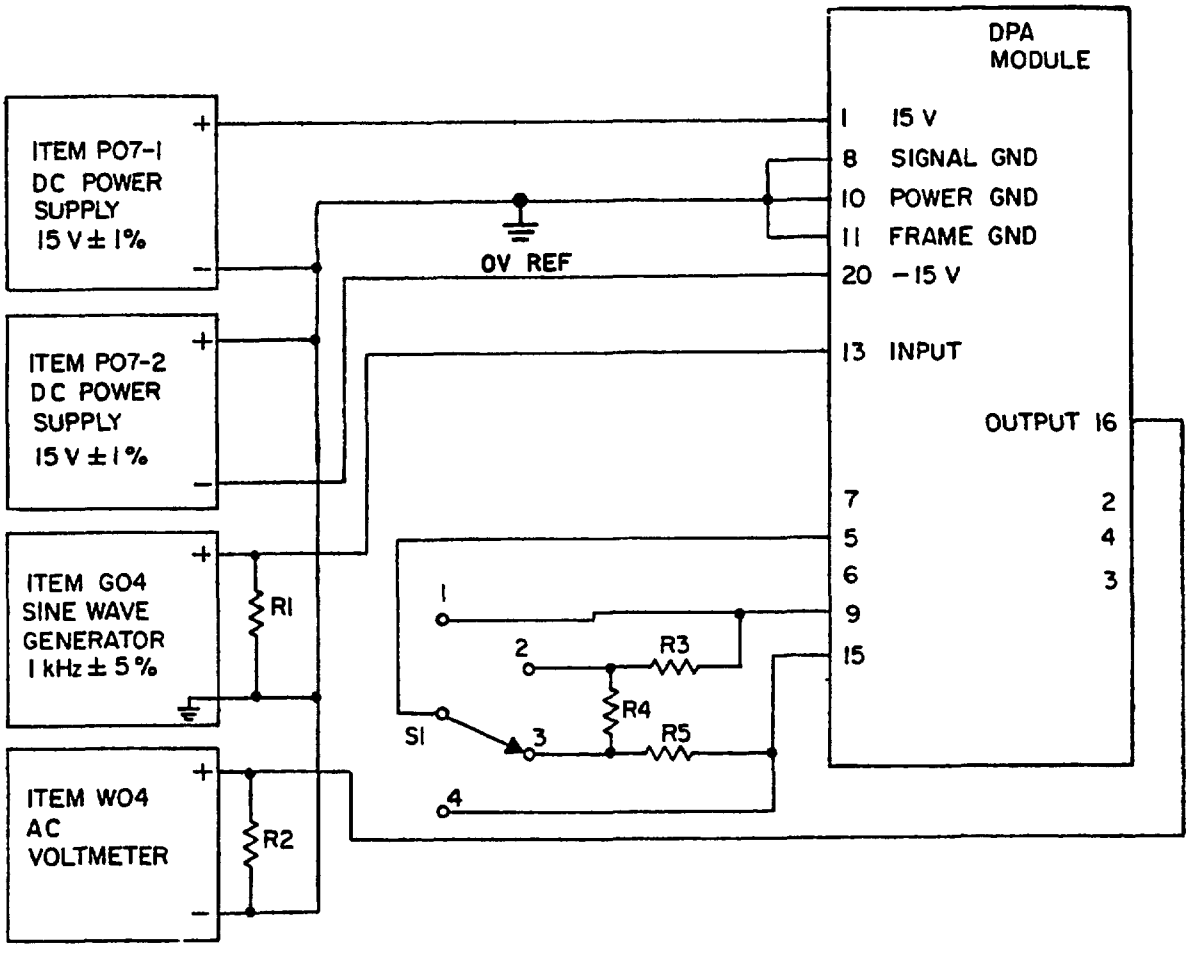
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NOTES:
1. R1 = R2 = 10 kΩ ±1%, 1/4 W.
2. S1 = SPDT switch.
3. With switch S1 in position 1, adjust item G01 until item W04 reads 100 mV rms ±1%, 1 kHz ±5%. Move switch S1 to position 2 and connect point A as specified in table XVI.

* FIGURE 18. Input impedance check test circuit for table XVI.

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NOTES:
1. R1 = 51Ω ±5%, 1 W.
2. R2 = 5 kΩ ±1%, 1/4 W.
3. R3 = R5 = 15 kΩ ±1%, 1/4 W.
4. R4 = 75 kΩ ±1%, 1/4 W.
5. S1 = SP, 4 position switch.
6. Test procedure:

   a. Record value of V1 for test condition R1.
   b. Use V1 as minimum limit for test condition R2.
   c. Record value of V2 for test condition R3.
   d. Use V2 as minimum limit for test condition R4.

FIGURE 19. Remote gain function test circuit for table XVII.

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5 PROBLEM AREAS

a Paragraph Number and Wording·

b Recommended Wording.

c Reason/Rationale for Recommendation

6. REMARKS

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